

Point

From Number Theory to Algebraic Geometry and Logic

Session P1 — Group Theory: Part I

Complete lecture notes with proofs, examples, and exercises

Abstract

This is the entry session of the course. It builds the language of *symmetry* — groups, subgroups, quotients, and actions — which will reappear as Galois groups (Act II), ideal class groups (S24), idele class groups (S49), fundamental groups (P3, S35), and automorphism groups of logical structures (Act VII). Every structural result used later is proved here: the subgroup criterion, Lagrange’s theorem and its corollaries, the injectivity criterion via kernels, the well-definedness of quotient groups, the three isomorphism theorems, and the Orbit–Stabilizer theorem. The session closes with the philosophical pattern the course will instantiate again and again: *a quotient is what remains when viewpoints that look the same are identified*.

Session plan (150 minutes)

Time	Block
0:00–0:10	Orientation: symmetry as the engine of the course; where groups return
0:10–0:30	§1 Groups: definition, examples, cancellation
0:30–0:55	§2 Subgroups; the criterion; cyclic groups; Lagrange and corollaries
0:55–1:05	<i>Break</i>
1:05–1:25	§3 Homomorphisms; kernels; injectivity; centers and inner automorphisms
1:25–1:55	§4 Normal subgroups; quotients; the three isomorphism theorems; three worked examples
1:55–2:20	§5 Actions; Orbit–Stabilizer; conjugation and the class equation; Cayley
2:20–2:30	Wrap-up: the quotient philosophy; bridge to P2–P3 and Act II

Conventions

Groups are written multiplicatively with identity e unless stated otherwise; $\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are additive unless a $\times$ or $*$ is displayed. For a subset S of a group, $S^{-1} = \{s^{-1} : s \in S\}$. All homomorphisms send e to e (proved, not assumed).

Contents

1	Groups: Definition and First Examples	3
2	Subgroups, Cyclic Groups, and Lagrange	3
3	Homomorphisms, Kernels, and Centers	5
4	Normal Subgroups, Quotients, and the Isomorphism Theorems	6
5	Group Actions, Orbits, and the Class Equation	7

6 Exercises	8
Where This Session Feeds the Course	9

1 Groups: Definition and First Examples

Definition 1.1 (Group). A **group** is a set G with a binary operation $G \times G \rightarrow G$, $(g, h) \mapsto gh$, such that:

- (G1) *Associativity*: $(gh)k = g(hk)$ for all $g, h, k \in G$;
- (G2) *Identity*: there exists $e \in G$ with $eg = ge = g$ for all g ;
- (G3) *Inverses*: for each g there exists g^{-1} with $gg^{-1} = g^{-1}g = e$.

The group is **abelian** if $gh = hg$ for all g, h . Its **order** is $|G|$ (possibly ∞).

Lemma 1.2 (Uniqueness and housekeeping). *In any group: the identity is unique; each inverse is unique; $(gh)^{-1} = h^{-1}g^{-1}$; $(g^{-1})^{-1} = g$; and $eg = g$ together with $hg = g$ already force $h = e$.*

Proof. If e, e' are identities then $e = ee' = e'$. If x, x' invert g then $x = xe = x(gx') = (xg)x' = ex' = x'$. Now $h^{-1}g^{-1}$ is an inverse of gh since $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = gg^{-1} = e$ and similarly on the other side; uniqueness gives $(gh)^{-1} = h^{-1}g^{-1}$. Taking $g = h^{-1}$ in $(gh)^{-1} = h^{-1}g^{-1}$ gives $(g^{-1})^{-1} = g$. Finally $hg = g = eg$ implies $h = e$ by left-cancelling g (multiply by g^{-1} on the right). $\square$

Proposition 1.3 (Cancellation laws). $gh = gk \Rightarrow h = k$ and $hg = kg \Rightarrow h = k$.

Proof. Multiply $gh = gk$ on the left by g^{-1} . $\square$

Example 1.4 (Running examples). • $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$: infinite abelian.

- $(\mathbb{Q}^*, \times)$, $(\mathbb{R}^*, \times)$, $(\mathbb{C}^*, \times)$: units; abelian.
- $(\mathbb{Z}/n\mathbb{Z}, +)$: finite abelian of order n .
- The **circle group** $U(1) = \{z \in \mathbb{C}^* : |z| = 1\}$ under multiplication; abelian; the group of rotations of the plane.
- The **symmetric group** $S_n = \text{Sym}(\{1, \dots, n\})$ under composition; non-abelian for $n \geq 3$; $|S_n| = n!$.
- $\text{GL}_n(k) = \{A \in M_n(k) : \det A \neq 0\}$ under multiplication; non-abelian for $n \geq 2$; $\text{SL}_n(k) = \{A : \det A = 1\}$; $\text{SO}_n(\mathbb{R}) = \{A \in \text{SL}_n(\mathbb{R}) : A^T A = I\}$ (rotations of $\mathbb{R}^n$).
- Direct products $G \times H$ with componentwise operation.
- The trivial group $\{e\}$.

Every new notion in this session should be tested on all of these.

Remark 1.5 (Why groups, why here). A group is “all the symmetries of an object, packaged”. The course will meet: symmetries of field extensions (Galois groups, Act II), symmetries of coverings (P3, S35), symmetries of lattices and curves (Acts III–V), and symmetries of logical theories (Act VII). The machinery below is the common grammar.

2 Subgroups, Cyclic Groups, and Lagrange

Definition 2.1 (Subgroup). $H \subseteq G$ is a **subgroup**, written $H \leq G$, if $e \in H$, H is closed under the operation, and H is closed under inverses. Equivalently H is a group with the restricted law.

Proposition 2.2 (Subgroup criterion). *A nonempty subset $H \subseteq G$ is a subgroup $\iff$ for all $x, y \in H$ one has $xy^{-1} \in H$.*

Proof. $(\Rightarrow)$ Clear. $(\Leftarrow)$ Take $x = y$: $e = xx^{-1} \in H$. Then for $y \in H$, $y^{-1} = ey^{-1} \in H$. Then for $x, y \in H$, $xy = x(y^{-1})^{-1} \in H$. So H satisfies the definition. $\square$

Example 2.3. $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$ (additive); $\text{SL}_n(k) \leq \text{GL}_n(k)$; $\text{SO}_n(\mathbb{R}) \leq \text{GL}_n(\mathbb{R})$; $U(1) \leq \mathbb{C}^*$; $n\mathbb{Z} \leq \mathbb{Z}$; and $\{e\} \leq G \leq G$ always. Non-example: $\mathbb{Z}_{\geq 0} \subseteq \mathbb{Z}$ (no inverses); $\{0, 2, 4\} \subseteq \mathbb{Z}/6\mathbb{Z}$ is a subgroup (check the criterion), of order 3.

Definition 2.4 (Generated and cyclic subgroups). For $S \subseteq G$, the **subgroup generated by S** is $\langle S \rangle = \bigcap \{H \leq G : S \subseteq H\}$, the smallest subgroup containing S . For one element, $\langle g \rangle$ makes G (or the subgroup) **cyclic** if $G = \langle g \rangle$ for some g ; such g is a **generator**.

Proposition 2.5. $\langle S \rangle$ equals the set of all finite products of elements of $S \cup S^{-1}$; in particular $\langle g \rangle = \{g^n : n \in \mathbb{Z}\}$, and every cyclic group is abelian.

Proof. The set of finite products of $S \cup S^{-1}$ contains S (length-one products), contains e (empty product), and is closed under product and inverse, hence is a subgroup containing S ; conversely any subgroup containing S contains all such products. For $S = \{g\}$ this gives $\{g^n\}$. Commutativity: $g^m g^n = g^{m+n} = g^n g^m$. $\square$

Definition 2.6 (Order of an element). $\text{ord}(g) = |\langle g \rangle|$; equivalently the smallest $n \geq 1$ with $g^n = e$, and ∞ if none.

Proposition 2.7. If $\text{ord}(g) = n < \infty$ then $g^k = e \iff n \mid k$, and $\langle g \rangle = \{e, g, \dots, g^{n-1}\}$ has exactly n elements. If $\text{ord}(g) = \infty$ then all g^k , $k \in \mathbb{Z}$, are distinct.

Proof. Write $k = qn + r$, $0 \leq r < n$; then $g^k = (g^n)^q g^r = g^r$, so $g^k = e \Rightarrow g^r = e \Rightarrow r = 0$ by minimality of n . Distinctness of $e, g, \dots, g^{n-1}$: if $g^i = g^j$ with $0 \leq j < i < n$ then $g^{i-j} = e$ with $0 < i - j < n$, contradiction. The infinite case is the same argument with no n . $\square$

Definition 2.8 (Cosets and index). For $H \leq G$ and $g \in G$, the **left coset** is $gH = \{gh : h \in H\}$; the set of left cosets is G/H ; the **index** is $[G : H] = |G/H|$.

Lemma 2.9. Left cosets partition G , and each has the same cardinality as H .

Proof. $g \in gH$ since $e \in H$, so cosets cover G . If $gH \cap g'H \neq \emptyset$ then $gh = g'h'$, so $g = g'h'h^{-1}$ and $gH \subseteq g'H$; symmetrically $g'H \subseteq gH$, so overlapping cosets coincide: a partition. The map $H \rightarrow gH$, $h \mapsto gh$, is bijective with inverse $x \mapsto g^{-1}x$. $\square$

Theorem 2.10 (Lagrange). If G is finite and $H \leq G$ then $|H|$ divides $|G|$, and $|G| = [G : H] \cdot |H|$.

Proof. G is the disjoint union of $[G : H]$ cosets, each of size $|H|$ (Lemma 2.9); sum the sizes. $\square$

Corollary 2.11. For finite G :

- (a) $\text{ord}(g)$ divides $|G|$ for every $g \in G$;
- (b) $g^{|G|} = e$ for every g ;
- (c) a group of prime order p is cyclic, generated by any $g \neq e$;
- (d) the only subgroups of $\mathbb{Z}/p\mathbb{Z}$ are $\{0\}$ and itself.

Proof. (a) $\text{ord}(g) = |\langle g \rangle|$ divides $|G|$ by Lagrange applied to $\langle g \rangle$. (b) $g^{|G|} = g^{\text{ord}(g) \cdot (|G|/\text{ord}(g))} = e$. (c) Any $g \neq e$ has $\text{ord}(g) \mid p$ and $\text{ord}(g) > 1$, so $\text{ord}(g) = p$ and $\langle g \rangle = G$. (d) Immediate from (c) and Lagrange. $\square$

Example 2.12. Subgroups of $\mathbb{Z}/12\mathbb{Z}$: one for each divisor, of orders 1, 2, 3, 4, 6, 12; element orders likewise divide 12. In S_3 (order 6): element orders are 1, 2, 3 only. In $\mathbb{F}_p^*$ (order $p - 1$): $a^{p-1} = 1$ for all $a \neq 0$ — this is Fermat's little theorem, obtained purely from Lagrange.

Remark 2.13 (The converse of Lagrange fails). A_4 has order 12 but *no* subgroup of order 6 (Exercise 9). The correct converse-direction statements are Sylow's theorems (mention; not needed later) and, in the course, the Galois correspondence (S13), which repairs the converse for *Galois* groups.

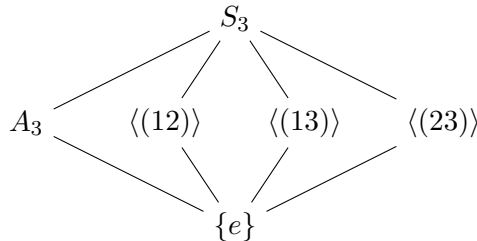


Figure 1: The subgroup lattice of S_3 : six subgroups, orders 1, 2, 2, 2, 3, 6 — all dividing 6, as Lagrange demands. Exactly one of the proper nontrivial ones (A_3) will turn out to be normal (§4).

3 Homomorphisms, Kernels, and Centers

Definition 3.1 (Homomorphism). $\varphi : G \rightarrow H$ is a **homomorphism** if $\varphi(gh) = \varphi(g)\varphi(h)$ for all g, h . Its **kernel** is $\ker \varphi = \{g \in G : \varphi(g) = e_H\}$ and its **image** $\text{im } \varphi = \varphi(G)$.

Proposition 3.2. $\varphi(e_G) = e_H$; $\varphi(g^{-1}) = \varphi(g)^{-1}$; $\ker \varphi \leq G$; $\text{im } \varphi \leq H$.

Proof. $\varphi(e) = \varphi(ee) = \varphi(e)\varphi(e)$, cancel $\varphi(e)$. Then $e = \varphi(e) = \varphi(gg^{-1}) = \varphi(g)\varphi(g^{-1})$ gives inverses. Subgroup criterion: $e \in \ker$; if $\varphi(x) = \varphi(y) = e$ then $\varphi(xy^{-1}) = ee = e$. Same for im with products and inverses in H . $\square$

Theorem 3.3 (Injectivity criterion). φ is injective $\iff \ker \varphi = \{e_G\}$.

Proof. $(\Rightarrow)$ $\varphi(e) = e$, so injectivity forces $\ker = \{e\}$. $(\Leftarrow)$ If $\varphi(g) = \varphi(h)$ then $e = \varphi(g)\varphi(h)^{-1} = \varphi(gh^{-1})$, so $gh^{-1} \in \ker = \{e\}$, hence $g = h$. $\square$

Definition 3.4 (Isomorphism; automorphism; center; inner automorphisms). An **isomorphism** is a bijective homomorphism ($G \cong H$). An **automorphism** is an isomorphism $G \rightarrow G$; they form $\text{Aut}(G)$. For $g \in G$, conjugation $c_g(x) = gxg^{-1}$ is an automorphism, called **inner**; the **center** is $Z(G) = \{g : gh = hg \ \forall h\}$.

Proposition 3.5. $c_g \in \text{Aut}(G)$; the map $g \mapsto c_g$ is a homomorphism $G \rightarrow \text{Aut}(G)$ with kernel $Z(G)$; hence $Z(G) \leq G$ and the inner automorphisms form a subgroup $\text{Inn}(G) \leq \text{Aut}(G)$.

Proof. $c_g(xy) = gxg^{-1}gyg^{-1} = c_g(x)c_g(y)$; bijective with inverse $c_{g^{-1}}$. $c_{gh} = c_g \circ c_h$ is direct. $c_g = \text{id} \iff gx = xg \ \forall x \iff g \in Z(G)$, so the kernel is $Z(G)$; kernels are subgroups (Proposition 3.2), and the image $\text{Inn}(G)$ is a subgroup of $\text{Aut}(G)$. $\square$

Example 3.6. • $\det : \text{GL}_n(k) \rightarrow k^*$ is a surjective homomorphism; $\ker \det = \text{SL}_n(k)$.

- **The sign homomorphism** $\text{sign} : S_n \rightarrow \{\pm 1\}$: let $\Delta = \prod_{i < j} (x_i - x_j) \in \mathbb{Z}[x_1, \dots, x_n]$; $\sigma \in S_n$ permutes the variables and sends each factor to $\pm$ a factor, so $\sigma(\Delta) = \pm \Delta$; define $\sigma(\Delta) = \text{sign}(\sigma)\Delta$. Then $(\sigma\tau)(\Delta) = \sigma(\tau(\Delta))$ gives multiplicativity; a transposition swaps one pair and fixes the parity of the rest, hence has sign -1 . Kernel = A_n (defined as $\ker$).
- $\exp : (\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \times)$ is an isomorphism with inverse $\log$.
- Reduction $\pi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is surjective with kernel $n\mathbb{Z}$.
- $Z(\text{GL}_n(k)) = \{\lambda I : \lambda \in k^*\}$; $Z(S_n) = \{e\}$ for $n \geq 3$ (exercises).

4 Normal Subgroups, Quotients, and the Isomorphism Theorems

Definition 4.1 (Normal subgroup). $N \leq G$ is **normal**, $N \trianglelefteq G$, if $gNg^{-1} = N$ for all $g \in G$.

Proposition 4.2 (Equivalent criteria). *For $N \leq G$ the following are equivalent:*

- (a) $gNg^{-1} = N$ for all g ;
- (b) $gNg^{-1} \subseteq N$ for all g ;
- (c) $gN = Ng$ for all g (left cosets = right cosets);
- (d) N is a union of conjugacy classes $\{gng^{-1} : n \in N\} \subseteq N$ pointwise-generated.

Proof. (a) $\Rightarrow$ (b) trivial. (b) $\Rightarrow$ (a): apply (b) to g^{-1} : $g^{-1}Ng \subseteq N$, so $N \subseteq gNg^{-1}$, and with (b) equality. (a) $\Leftrightarrow$ (c): $gN = Ng \iff gNg^{-1} = N$ by multiplying on the right by g^{-1} . (d) is a restatement of (b). $\square$

Proposition 4.3 (Kernels are normal; and only kernels). $\ker \varphi \trianglelefteq G$ for any homomorphism φ ; conversely every normal subgroup is the kernel of the projection constructed in Theorem 4.5.

Proof. $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g)^{-1} = e$, so $g \ker \varphi g^{-1} \subseteq \ker \varphi$; apply Proposition 4.2(b). $\square$

Example 4.4. In abelian groups every subgroup is normal. $A_n \trianglelefteq S_n$ and $\mathrm{SL}_n \trianglelefteq \mathrm{GL}_n$ as kernels. $\mathbb{Z}(G) \trianglelefteq G$. Non-example: $\langle (12) \rangle \leq S_3$ is *not* normal: $(13)(12)(13)^{-1} = (23) \notin \langle (12) \rangle$ — compare Figure 1: normality is rare.

Theorem 4.5 (Quotient groups). *Let $N \trianglelefteq G$. On the coset set G/N the rule*

$$(gN) \cdot (hN) := (gh)N$$

is well defined and makes G/N a group; the projection $\pi : G \rightarrow G/N$, $g \mapsto gN$, is a surjective homomorphism with $\ker \pi = N$; and $|G/N| = [G : N]$.

Proof. Well-definedness: if $g' = gn_1$, $h' = hn_2$ then $g'h'N = gn_1hn_2N = gh(h^{-1}n_1h)n_2N = ghN$ since $h^{-1}n_1h \in N$ by normality and $NN = N$. Associativity inherits from G ; identity $eN = N$; inverse $(gN)^{-1} = g^{-1}N$. π is a homomorphism by the very definition of the product; $\pi(g) = N \iff gN = N \iff g \in N$. $\square$

Theorem 4.6 (First Isomorphism Theorem). *For $\varphi : G \rightarrow H$ a homomorphism, the map*

$$\bar{\varphi} : G/\ker \varphi \longrightarrow \mathrm{im} \varphi, \quad g \ker \varphi \longmapsto \varphi(g)$$

is a well-defined isomorphism; in particular $G/\ker \varphi \cong \mathrm{im} \varphi$.

Proof. Well-defined: if $g' = gk$, $k \in \ker$, then $\varphi(g') = \varphi(g)\varphi(k) = \varphi(g)$. Homomorphism: inherited. Injective: $\bar{\varphi}(g \ker) = e \Rightarrow \varphi(g) = e \Rightarrow g \ker = \ker$. Surjective onto $\mathrm{im} \varphi$ by definition. $\square$

Example 4.7 (Three worked applications). (i) $\det : \mathrm{GL}_n(k) \twoheadrightarrow k^*$, $\ker = \mathrm{SL}_n(k) \Rightarrow \mathrm{GL}_n(k)/\mathrm{SL}_n(k) \cong k^*$. Over $k = \mathbb{F}_q$: $|\mathrm{SL}_n(\mathbb{F}_q)| = |\mathrm{GL}_n(\mathbb{F}_q)|/(q-1)$.

(ii) $\mathrm{sign} : S_n \twoheadrightarrow \{\pm 1\}$, $\ker = A_n \Rightarrow S_n/A_n \cong \mathbb{Z}/2$, hence $|A_n| = n!/2$.

(iii) $\pi : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/n\mathbb{Z}$, $\ker = n\mathbb{Z} \Rightarrow \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z}$ (tautological but canonical); more usefully: any homomorphism $\mathbb{Z} \rightarrow G$ is $1 \mapsto g$, kernel $\mathrm{ord}(g)\mathbb{Z}$, so $\mathbb{Z}/\mathrm{ord}(g)\mathbb{Z} \cong \langle g \rangle$ — every cyclic group is a quotient of $\mathbb{Z}$.

Theorem 4.8 (Second and Third Isomorphism Theorems). *Let G be a group.*

(a) (Second) If $H \leq G$ and $N \trianglelefteq G$ then $HN = \{hn\} \leq G$, $N \trianglelefteq HN$, $H \cap N \trianglelefteq H$, and $HN/N \cong H/(H \cap N)$.

(b) (Third) If $N \trianglelefteq G$ and $N \subseteq H \leq G$ then $H/N \trianglelefteq G/N$ and $(G/N)/(H/N) \cong G/H$.

Proof. (a) HN is a subgroup because N normal: $(h_1n_1)(h_2n_2) = h_1h_2(h_2^{-1}n_1h_2)n_2 \in HN$, and inverses similarly. Define $\psi : H \rightarrow HN/N$ by $\psi(h) = hN$; this is a homomorphism (restriction of π), surjective since $hnN = hN$, and $\ker \psi = \{h : h \in N\} = H \cap N$; apply Theorem 4.6. (b) The map $G/N \rightarrow G/H$, $gN \mapsto gH$, is a well-defined surjective homomorphism (well-defined since $N \subseteq H$) with kernel $\{gN : g \in H\} = H/N$; apply Theorem 4.6. $\square$

Remark 4.9 (The quotient philosophy). G/N is what you see of G when you declare everything in N invisible. The First Isomorphism Theorem says: *every* homomorphic image of G arises this way, and the kernel records exactly what was blurred. This is the group-theoretic prototype of: norm groups blurring ideles in Artin reciprocity (S47–S49), residue fields blurring local rings (S1, S28), and models identifying provably equal terms (Act VII). Quotients are *spaces of viewpoints collapsed to points*.

5 Group Actions, Orbits, and the Class Equation

Definition 5.1 (Action). A **(left) action** of G on a set X is a map $G \times X \rightarrow X$, $(g, x) \mapsto g \cdot x$, with $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$.

Proposition 5.2. *Actions of G on X are in bijection with homomorphisms $\rho : G \rightarrow \text{Sym}(X)$, via $\rho(g)(x) = g \cdot x$.*

Proof. Given an action, each $\rho(g)$ is bijective with inverse $\rho(g^{-1})$, and $\rho(gh) = \rho(g)\rho(h)$ is the action axiom; so ρ is a homomorphism. Conversely a homomorphism defines an action by $g \cdot x := \rho(g)(x)$; the two constructions invert each other. $\square$

Example 5.3. • S_n on $\{1, \dots, n\}$ (tautological); on k -element subsets; on ordered tuples.

- $\text{GL}_n(k)$ on k^n by matrix multiplication; on lines through the origin (projective space).
- **Conjugation:** G on itself, $g \cdot x = gxg^{-1}$; and on its subgroups, $g \cdot H = gHg^{-1}$ (normal subgroups = fixed points of this action).
- **Left multiplication:** G on itself, $g \cdot x = gx$ (free and transitive).
- D_4 (symmetries of the square) on its four vertices; rotation subgroup on vertices.
- Trivial action $g \cdot x = x$ for all g .

Definition 5.4 (Orbit, stabilizer, fixed points). $\text{Orb}(x) = \{g \cdot x : g \in G\}$; $\text{Stab}(x) = \{g : g \cdot x = x\} \leq G$; $X^G = \{x : g \cdot x = x \ \forall g\}$. The action is **transitive** if there is one orbit, **free** if all stabilizers are trivial.

Theorem 5.5 (Orbit–Stabilizer). *The map $G/\text{Stab}(x) \rightarrow \text{Orb}(x)$, $g\text{Stab}(x) \mapsto g \cdot x$, is a well-defined bijection; hence for finite G ,*

$$|G| = |\text{Orb}(x)| \cdot |\text{Stab}(x)|.$$

Proof. Well-defined: $g' = gs$ with $s \in \text{Stab}(x)$ gives $g' \cdot x = g \cdot (s \cdot x) = g \cdot x$. Injective: $g \cdot x = g' \cdot x \Rightarrow g^{-1}g' \cdot x = x \Rightarrow g^{-1}g' \in \text{Stab}(x) \Rightarrow g\text{Stab} = g'\text{Stab}$. Surjective: by definition of orbit. Counting: $|G/\text{Stab}(x)| = [G : \text{Stab}(x)] = |G|/|\text{Stab}(x)|$ by Lagrange. $\square$

Example 5.6 (Computing orders via Orbit–Stabilizer). • S_n on $\{1, \dots, n\}$: orbit of 1 has size n ; $\text{Stab}(1) \cong S_{n-1}$; inductively $|S_n| = n \cdot (n-1)!$ — Lagrange-free derivation of $n!$.

- $\mathrm{GL}_n(\mathbb{F}_q)$ on $\mathbb{F}_q^n \setminus \{0\}$: orbit is everything (transitive), stabilizer of e_1 has $q^{n-1}(q^{n-1}-1)\cdots$ — cleanly: $|\mathrm{GL}_n(\mathbb{F}_q)| = (q^n-1)(q^n-q)\cdots(q^n-q^{n-1})$ by choosing columns (exercise, both ways).
- D_4 on vertices: orbit size 4, stabilizer of a vertex has order 2, so $|D_4| = 8$.

Proposition 5.7 (Class equation). *For finite G , partitioning G into conjugacy classes (orbits of the conjugation action), with representatives x_i of the non-singleton classes:*

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)], \quad C_G(x) = \mathrm{Stab}_{\mathrm{conj}}(x) = \{g : gx = xg\}.$$

Proof. Orbits of the conjugation action partition G ; singleton orbits are exactly central elements; non-singleton orbits have size $[G : C_G(x_i)]$ by Orbit–Stabilizer. Sum orbit sizes. $\square$

Theorem 5.8 (Cayley). *Every group G embeds into $\mathrm{Sym}(G)$; in particular every finite group of order n is isomorphic to a subgroup of S_n .*

Proof. The left-multiplication action gives $\rho : G \rightarrow \mathrm{Sym}(G)$ (Proposition 5.2); $\ker \rho = \{g : gx = x \forall x\} = \{e\}$ (take $x = e$), so ρ is injective by Theorem 3.3. $\square$

Remark 5.9 (Where actions return). Galois groups act on roots and on intermediate fields (S12–S13); π_1 acts on fibers of coverings (P3, S35); Frobenius elements act on points of varieties mod p (S9, S40, S46); automorphism groups act on models of theories (Act VII). Orbit–Stabilizer is the counting engine behind all of these, and the class equation is the seed of the simplicity proof for A_n (S11).

6 Exercises

Exercise 1. Prove that the intersection of any family of subgroups is a subgroup, and that $\langle S \rangle$ as defined by intersection equals the set of finite products of $S \cup S^{-1}$.

Exercise 2. List all subgroups of $\mathbb{Z}/12\mathbb{Z}$ and of S_3 ; draw both lattices; identify which are cyclic and which are normal.

Exercise 3. Show that a nonempty finite subset H of a group closed under the operation is already a subgroup (finiteness is essential: find a counterexample in $(\mathbb{Z}, +)$).

Exercise 4. Prove: if G is a group in which $g^2 = e$ for all g , then G is abelian. More generally, if $(gh)^2 = g^2h^2$ for all g, h , then G is abelian.

Exercise 5. Prove sign as defined in Example 3.6 is a well-defined homomorphism and that a transposition has sign -1 . Conclude $|A_n| = n!/2$.

Exercise 6. Show $Z(\mathrm{GL}_n(k)) = \{\lambda I : \lambda \in k^*\}$ and $Z(S_n) = \{e\}$ for $n \geq 3$.

Exercise 7. Give an explicit homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ with $\mathrm{im} \varphi = 6\mathbb{Z}$; then describe $\mathbb{Z}/\ker \varphi$ and check the First Isomorphism Theorem by hand.

Exercise 8. In S_4 , take $H = \langle (12) \rangle$ and $N = \{e, (12)(34), (13)(24), (14)(23)\}$ (check $N \trianglelefteq S_4$). Verify the Second Isomorphism Theorem: compute HN , $H \cap N$, and both sides of $HN/N \cong H/(H \cap N)$.

Exercise 9. Prove A_4 has no subgroup of order 6 (hint: such a subgroup would be normal of index 2; look at the three double transpositions). This shows the converse of Lagrange fails.

Exercise 10. Using Orbit–Stabilizer, compute $|\mathrm{GL}_n(\mathbb{F}_q)|$ in two ways: (i) by choosing a basis column by column; (ii) via the action on $\mathbb{F}_q^n \setminus \{0\}$ and induction on the stabilizer.

Exercise 11. Write the class equation of S_4 explicitly (conjugacy classes = cycle types) and verify the sum.

Exercise 12. Let G act on X . Prove that orbits partition X , and that the action is transitive $\iff X \cong G/\text{Stab}(x)$ as G -sets for any x .

Where This Session Feeds the Course

- **Depends on:** nothing — this is the entry point of the course.
- **Feeds:** Act II entirely (Galois groups, simplicity of A_n via the class equation, solvability); S24 (ideal class group as a quotient group); S47–S49 (Artin reciprocity as a First Isomorphism Theorem at global scale: $C_K/N(C_L) \cong \text{Gal}(L/K)$); P3 and S35 (deck transformations and étale π_1 as permutation groups of fibers); Act VII (automorphism groups of models).
- **Unifying thread:** a quotient group is a space of viewpoints collapsed to points; a group action is a family of points moving under symmetry; kernels record exactly what an evaluation (homomorphism) cannot see.

References for this session

- [1] Artin, M. *Algebra*, ch.2.
- [2] Herstein, I.N. *Topics in Algebra*, ch.2.
- [3] Rotman, J.J. *An Introduction to the Theory of Groups*, ch.1–4.
- [4] Dummit, D. & Foote, R. *Abstract Algebra*, ch.1–4 (supplementary, rich exercises).
- [5] Lang, S. *Algebra*, ch.I (supplementary).

End of Session P1 — next: Session P2, Topology and Metric Spaces